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(6L)

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Midsemester Exam #4

Only one sheet of paper 8 1/2 x 11 is allowed

No phones or i-pads or computers are allowed

Duration: 75 minutes --- December 4, 2025

All Problems must be done

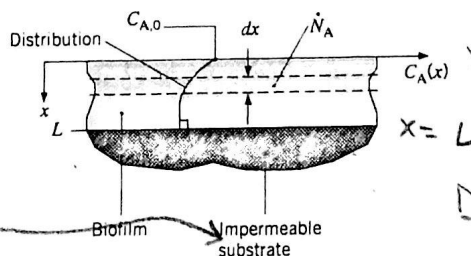
Problem 1: (35 points)

Biofilms, which are colonies of bacteria that can cling to living or inert surfaces, can cause a wide array of human infections. Infections caused by bacteria living within biofilms are often chronic because antibiotics that are applied to the surface of a biofilm have difficulty penetrating through the film thickness. Consider a biofilm that is associated with a skin infection. An antibiotic (species A) is applied to the top layer of a biofilm (species B) so that a fixed concentration of medication, $C_{A,0} = 4 \times 10^{-3} \text{ kmol/m}^3$, exists at the upper surface of the biofilm. The diffusion coefficient of the medication within the biofilm is $D_{AB} = 2 \times 10^{-12} \text{ m}^2/\text{s}$. The antibiotic is consumed by biochemical reactions within the film, and the consumption rate depends on the local concentration of medication expressed as $\dot{N}_A = -k_1 C_A$ where $k_1 = 0.1 \text{ s}^{-1}$. To eradicate the bacteria, the antibiotic must be consumed at a rate of at least $0.2 \times 10^{-3} \text{ kmol/s} \cdot \text{m}^3$ ($\dot{N}_A \leq -0.2 \times 10^{-3} \text{ kmol/s} \cdot \text{m}^3$) since, at smaller absolute consumption rates, the bacteria will be able to grow back faster than it is destroyed. Determine the maximum thickness of a biofilm, L , that may be treated successfully by the antibiotic.

$$k_1 \rightarrow \frac{1}{\text{s}} \quad \frac{1}{\text{s} \cdot \frac{\text{m}^2}{\text{s}}} \rightarrow \frac{1}{\text{m}^2} \rightarrow \frac{1}{\text{m}}$$



Schematic:



$$D \frac{d^2 C_A}{dx^2} - k_1 C_A + k_b = 0$$

$$\downarrow$$

$$D r^2 - k_1 = 0 \rightarrow r = \pm \sqrt{\frac{k_1}{D}}$$

$$r = \pm 2.24 \times 10^5 \text{ m}^{-1}$$

$$D \frac{d^2 C_A}{dx^2} - k_1 C_A + k_b = 0$$

$$-k_1 C_A = -k_b \rightarrow C_A = \frac{k_b}{k_1}$$

$$0.2 \times 10^{-3} = 2.24 \times 10^5 D e^{2.24 \times 10^5 L} + E e^{-2.24 \times 10^5 L}$$

$$C_A = D e^{2.24 \times 10^5 x} + E e^{-2.24 \times 10^5 x}$$

$$C_{A,0} = D + E$$

$$C_A(x=0) = C_{A,0}$$

$$D \left(\frac{dC_A}{dx} \right)_{x=0} = 0$$

$$-D \left(\frac{dC_A}{dx} \right)_{x=L} = 0.2 \times 10^{-3}$$

$$C_A(x=L) = 0.2 \times 10^{-3}$$

$$50 \mu\text{m} \rightarrow .05 \text{ mm}$$

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Problem 2: (35 points)

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The efficacy of pharmaceutical products is reduced by prolonged exposure to high temperature, light, and humidity. For water vapor-sensitive consumer products that are in tablet or capsule form, and might be stored in humid environments such as bathroom medicine cabinets, *blister packaging* is used to limit the direct exposure of the medicine to humid conditions until immediately before its ingestion.

Consider tablets that are contained in a blister package composed of a flat *lidding sheet* and a second, *formed sheet* that includes troughs to hold each tablet. The formed sheet is $L = 50 \mu\text{m}$ thick and is fabricated of a polymer material. Each trough is of diameter $D = 5 \text{ mm}$ and depth $h = 3 \text{ mm}$. The lidding sheet is fabricated of aluminum foil. The binary diffusion coefficient for water vapor in the polymer is $D_{AB} = 6 \times 10^{-14} \text{ m}^2/\text{s}$ while the aluminum may be assumed to be impermeable to water vapor. For molar concentrations of water vapor in the polymer at the outer and inner surfaces of $C_{A,s1} = 4.5 \times 10^{-3} \text{ kmol/m}^3$ and $C_{A,s2} = 0.5 \times 10^{-3} \text{ kmol/m}^3$, respectively, determine the rate at which water vapor is transferred through the trough wall to the tablet.

$$\text{Flux: } \frac{C_A}{A}$$

or

$$-D \left(\frac{dC_A}{dx} \right) \cdot A$$

BC

$$C_A(x=0) = C_{A,s1}$$

$$C_A(x=L) = C_{A,s2}$$

$$-D_{AL} \left(\frac{dC_A}{dx} \right) \Big|_{x=L} = 0$$

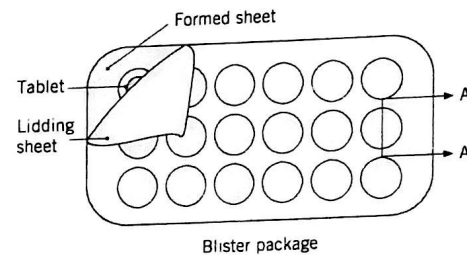
Lidding sheet

$$D_{AB} \frac{d^2 C_A}{dx^2} = 0$$

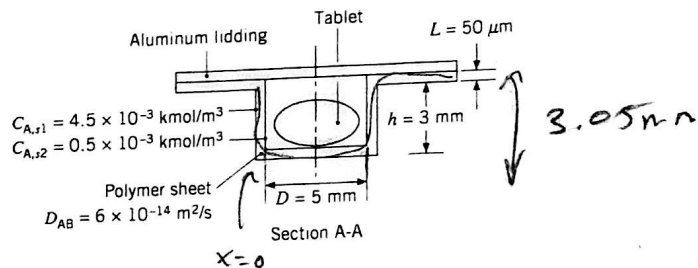
$$C_A = E x + F$$

Polymer.

$$D_{AB} \frac{d^2 C_A}{dx^2} = 0$$



Schematic:



thick

$$L = 50 \mu\text{m} \rightarrow .05 \text{ mm}$$

diameter

$$D = 5 \text{ mm} \rightarrow 2.5 \text{ mm}$$

height

$$h = 3 \text{ mm}$$

$$A = D \cdot h \rightarrow 15 \text{ mm}^2$$

$$C_{A,s1} = F$$

$$C_{A,s2} = E (L = .05 \text{ mm}) + C_{A,s1}$$

$$E = \frac{-0.004 \text{ kmol/m}^3}{.05 \text{ mm} \rightarrow .005 \text{ m}} = -8 \frac{\text{kmol}}{\text{m}^4}$$

$$C_A = -8 \frac{\text{kmol}}{\text{m}^4} x + 4.5 \times 10^{-3} \frac{\text{kmol}}{\text{m}^3}$$

$$N_A = -D_{AB} \left(-8 \frac{\text{kmol}}{\text{m}^4} \right) \cdot 15 \text{ mm}^2 \rightarrow 15 \text{ mm}^2 \cdot \left(\frac{1 \text{ m}}{1000 \text{ mm}} \right)^2 = 1.5 \times 10^{-5} \text{ m}^2$$

$$6 \times 10^{-14} \frac{\text{m}^2}{\text{s}} \left(8 \frac{\text{kmol}}{\text{m}^4} \right) \cdot 1.5 \times 10^{-5} \text{ m}^2 = 7.2 \times 10^{-19} \text{ kmol}$$

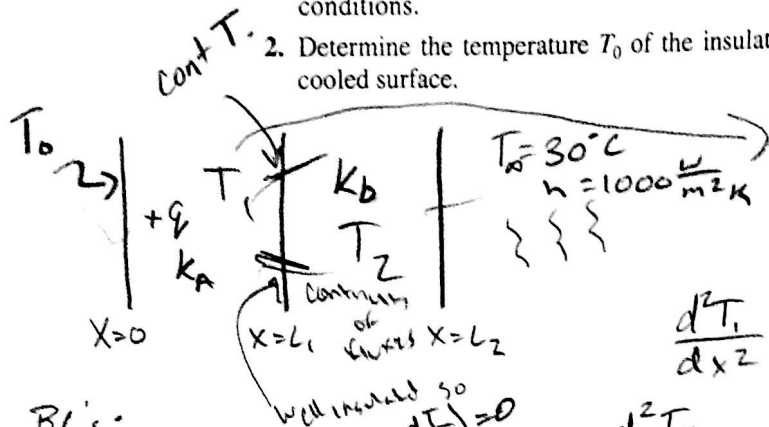
makes sense since we want to keep humidity out

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Problem 3: (35 points)

A plane wall is a composite of two materials, A and B. The wall of material A has uniform heat generation $\dot{q} = 1.5 \times 10^6 \text{ W/m}^3$, $k_A = 75 \text{ W/m} \cdot \text{K}$, and thickness $L_A = 50 \text{ mm}$. The wall material B has no generation with $k_B = 150 \text{ W/m} \cdot \text{K}$ and thickness $L_B = 20 \text{ mm}$. The inner surface of material A is well insulated, while the outer surface of material B is cooled by a water stream with $T_\infty = 30^\circ\text{C}$ and $h = 1000 \text{ W/m}^2 \cdot \text{K}$.

1. Sketch the temperature distribution that exists in the composite under steady-state conditions.
2. Determine the temperature T_0 of the insulated surface and the temperature T_2 of the cooled surface.



$$K \frac{d^2 T_1}{dx^2} + q = 0$$

$$K_b \frac{d^2 T_2}{dx^2} = 0$$

$$\frac{d^2 T_1}{dx^2} = -\frac{q}{k_A} \rightarrow T_1 = -\frac{q}{2k_A} x^2 + Ax + B$$

$$\frac{d^2 T_2}{dx^2} \Rightarrow T_2 = Cx + D$$

$$T_1(x=0) \rightarrow T_0 = B \quad \text{so} \quad T_1 = -\frac{q}{2k_p} x^2 + Ax + T_0$$

$$T_2(x=L_1+L_2) \rightarrow T_{2S} = C(L_1+L_2) + D$$

$$-k_A \left(\frac{dT_1}{dx} \right) = -k_B \left(\frac{dT_2}{dx} \right) \Big|_{x=L_1} \quad T_2(x=L_1+L_2) \rightarrow T_{2s} = C(L_1+L_2) + D$$

$$-k_B \left(\frac{dT_2}{dx} \right) = h(T - T_\infty) \Big|_{x=L_1+L_2} \quad -k_A \left(-\frac{q}{k_A} L_1 + A \right) = -k_B(C)$$

$$-k_b(C) = h(T - T_{\infty}) \rightarrow C = \frac{h(T - T_{\infty})}{-k_b}$$

$$T_1 = T_2|_{x=L_1} \quad -\frac{q}{2k_A}x^2 + A_x + T_0 = C_x + D \rightarrow -\frac{q}{2k_A}L_1^2 + AL_1 + T_0 = CL_1 + D$$

$$T_1(x=0) = T_0 \rightarrow T_0 = B \rightarrow -\frac{q}{2k_A} x^2 + Ax + T_0$$

$$T_z(x_{s,1}, \dot{q}) = h(T - T_\infty) \rightarrow -C(L_1 + L_2) + D = h(T_2 - T_\infty)$$

$$-k_A \left(\frac{dT}{dx} \right) = k_B \left(\frac{dT}{dx} \right) \Big|_{x=L_1} \rightarrow -k_A \left(-\frac{q}{k_A} L_1 + A \right) = -k_B (0) \rightarrow C = \frac{k_A}{k_B} \left(-\frac{q}{k_A} L_1 + A \right)$$

$$T_1 = T_2|_{x=L_1} \rightarrow -\frac{q}{2k_A} L_1^2 + A L_1 + T_0 = C L_1 + D$$